FAKULTI TEKNOLOGI DAN KEJURUTERAAN ELEKTRONIK & KEJURUTERAAN KOMPUTER

ASSIGNMENT

BERR 2243 Database and Cloud System

SEMESTER 2 2025/2026

NO.	NAME	MATRICS NUMBER
1.	MUHAMMAD HAIKAL BIN MOHD NIZAR	B122510461
2.	MUHAMMAD SOFWAN BIN MOHAMAD SABRI	B122510458
3.	MUHAMMAD AMMAR BIN ZINE BAZURI	B122510473
4.	ARIF AFFANDI BIN SHAHIDAN	B122510453

LECTURER: Professor Madya Dr Soo Yew Guan

1. Introduction

The objective of this project is to design and implement a complete end-to-end IoT cloud system integrating an ESP32 microcontroller, EMQX MQTT broker, Node-RED middleware, MongoDB Atlas database, and MongoDB Charts dashboard. The system enables real-time collection, processing, storage, and visualization of environmental sensor data using a secure cloud architecture.

2. System Architecture Overview

The system consists of five main layers: Edge Layer (ESP32), Cloud Middleware (EMQX and Node-RED), Database Layer (MongoDB Atlas), Security Layer, and Application Layer (MongoDB Charts). Data is transmitted using MQTT protocol in JSON format from ESP32 to cloud services.

Edge Layer: ESP32 publishes temperature and humidity data via MQTT.

Cloud Layer: EMQX handles MQTT messaging; Node-RED processes and validates data.

Database Layer: MongoDB Atlas stores metadata and time-series sensor data.

Security Layer: RBAC and authentication restrict unauthorized access.

Application Layer: MongoDB Charts visualizes real-time and historical data.

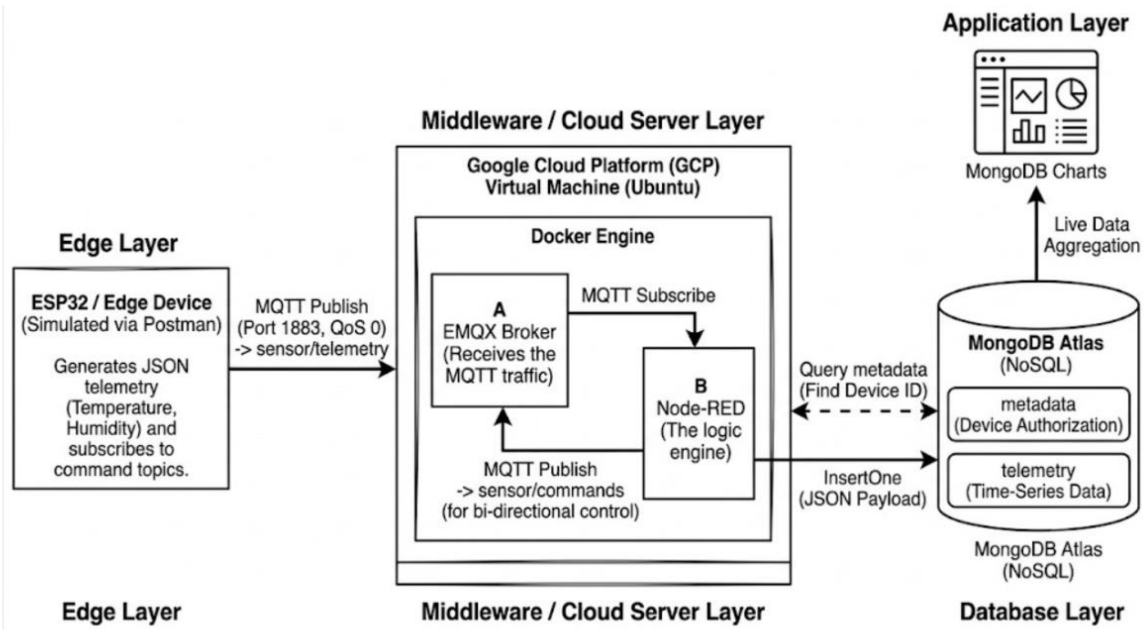
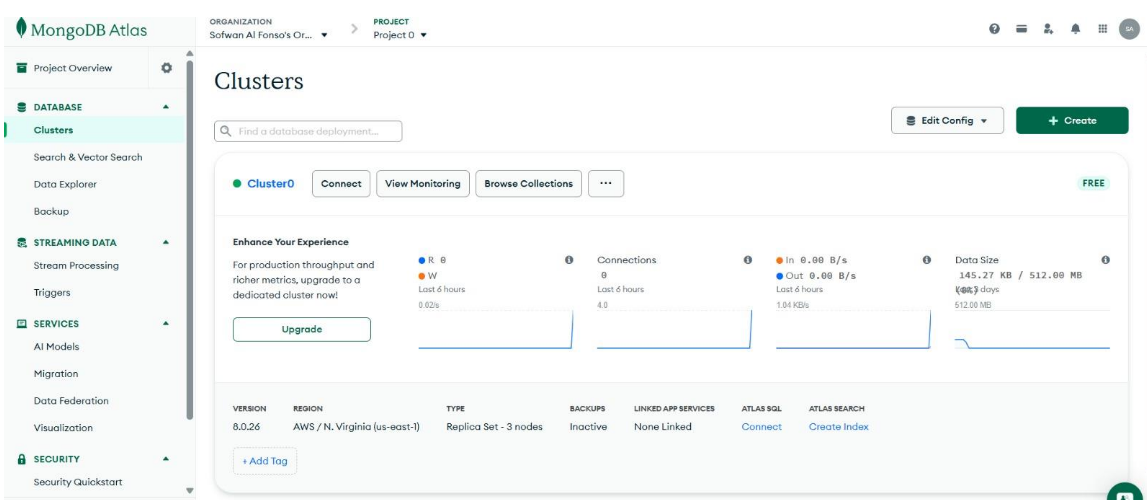


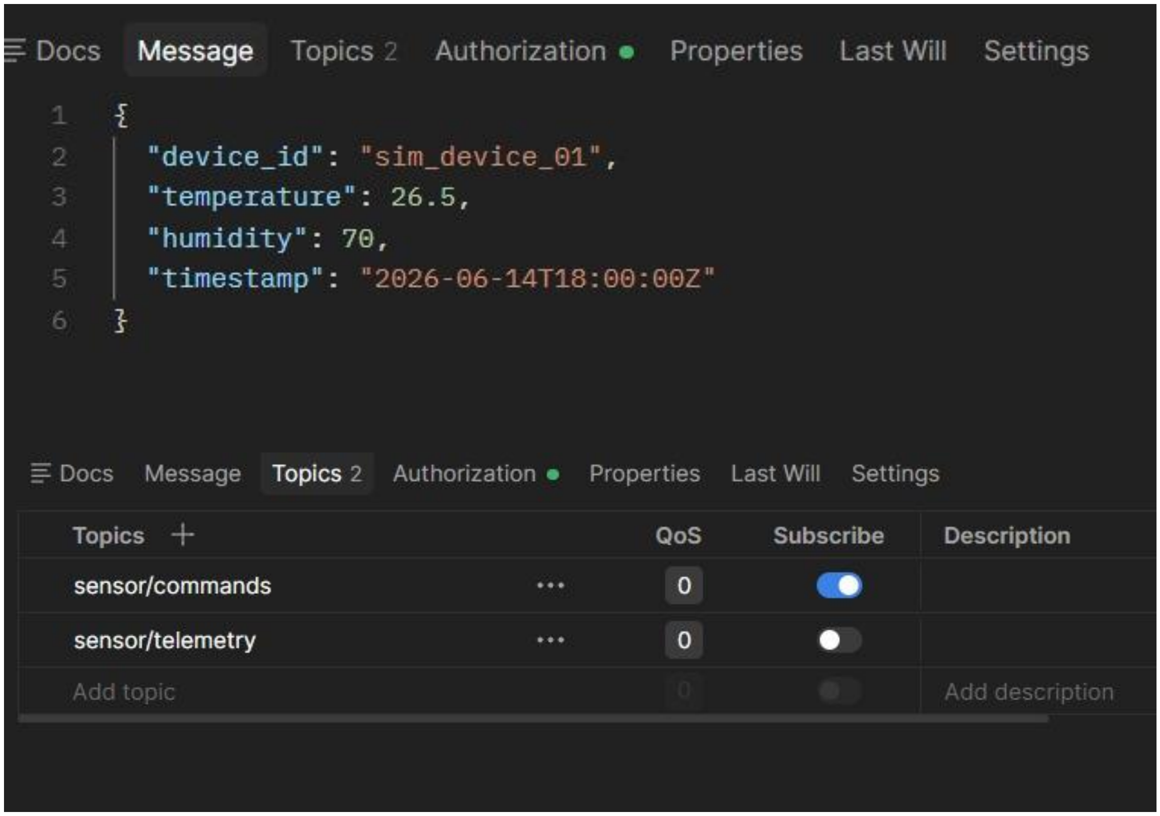
Figure 1: System Architecture

3. Deployment Process

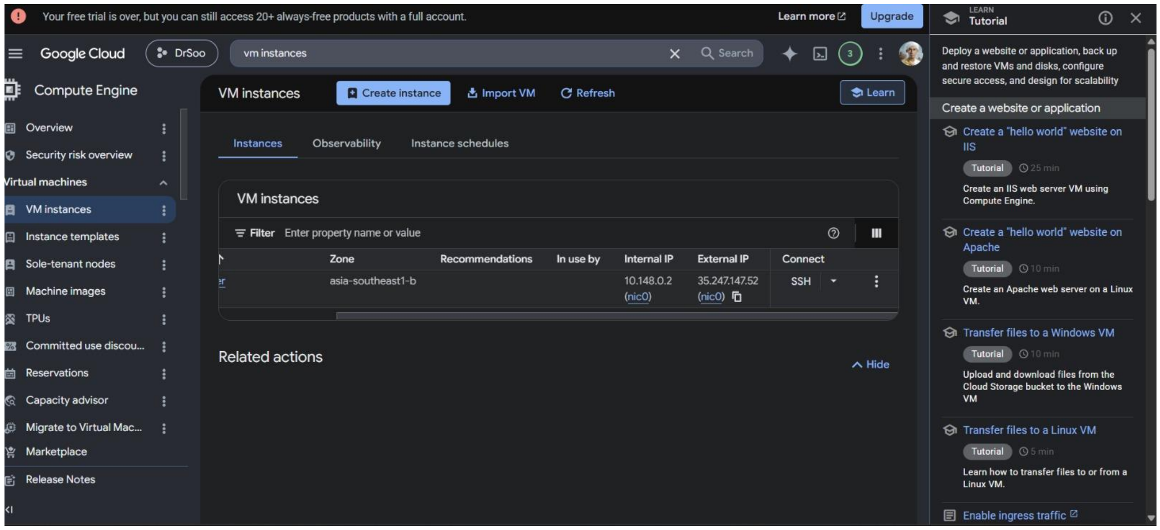
3.1 Create cluster 0 at MongoDB Atlas

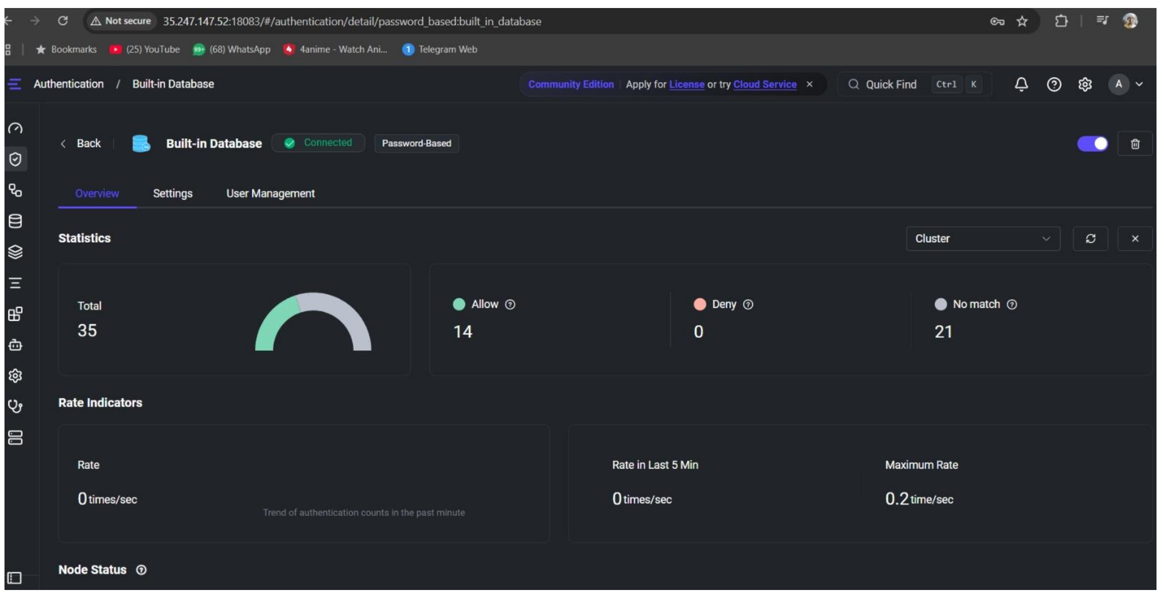


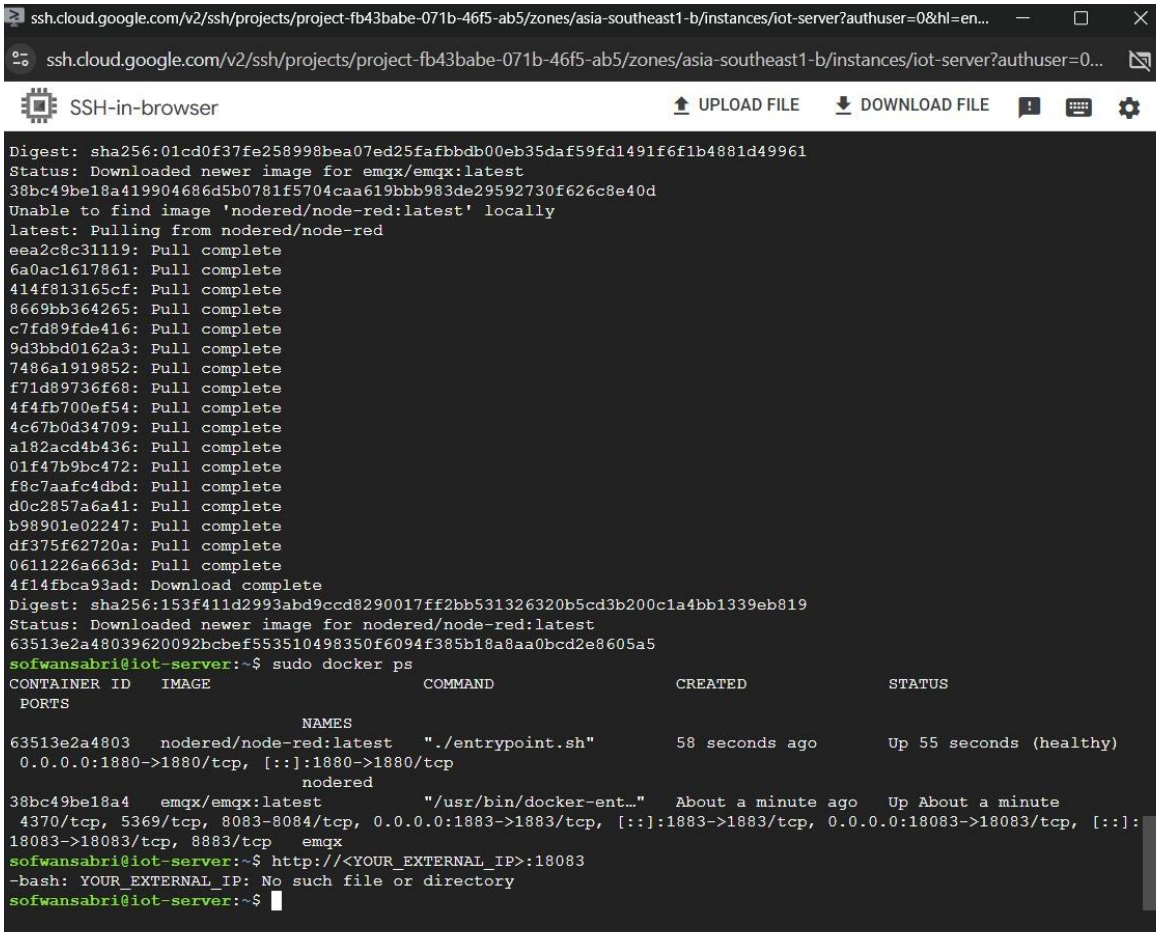
3.2 ESP32 programmed using postman to publish JSON sensor data.



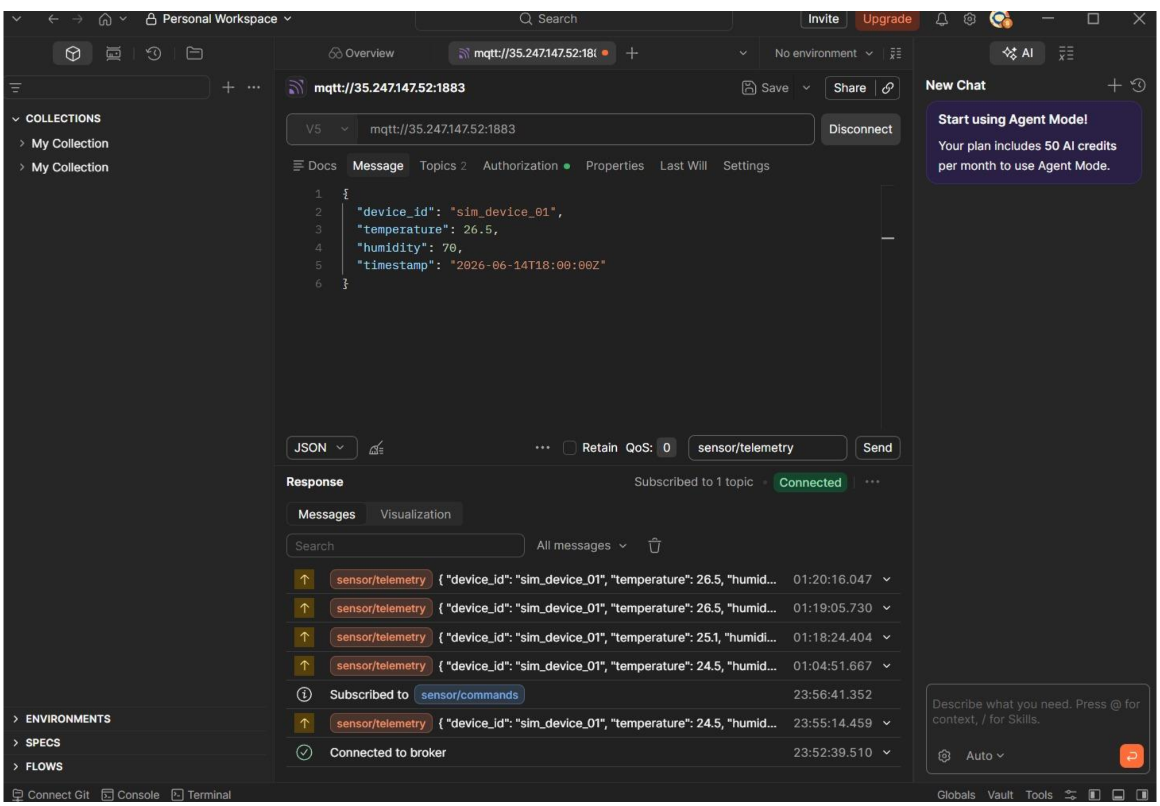
3.3 Cloud VM deployed with EMQX and Node-RED using Google Cloud.

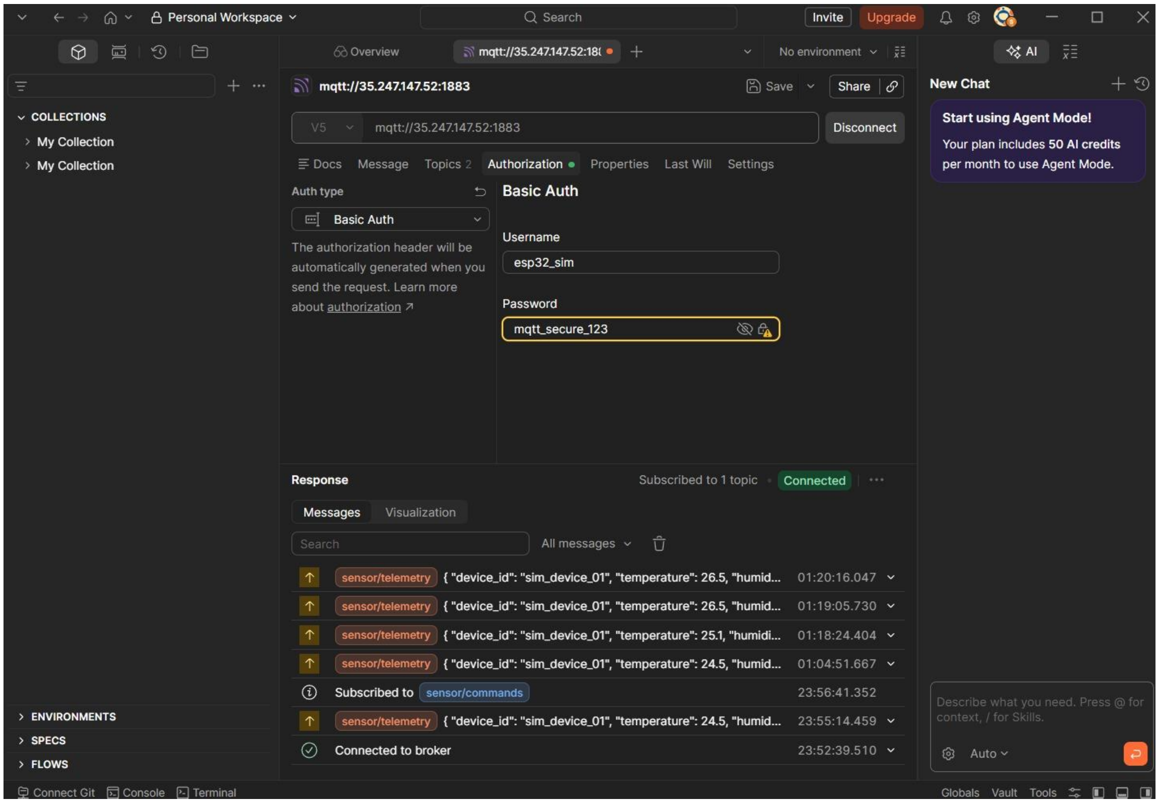




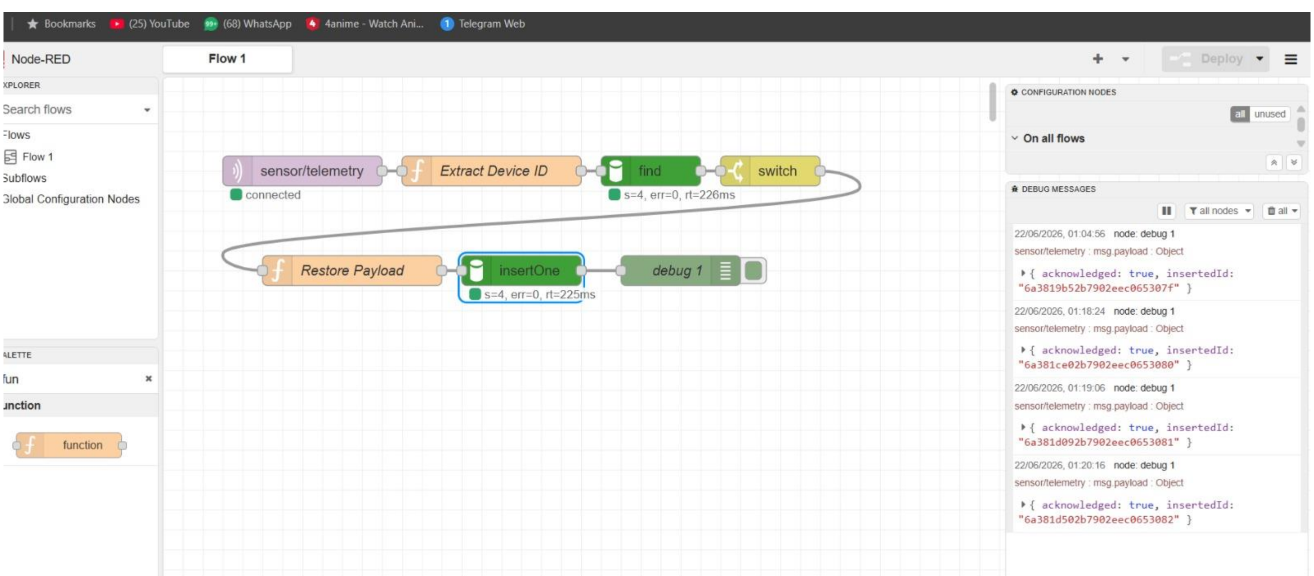


3.4 MQTT communication established between ESP32(postman) and EMQX broker.





3.5 Node-RED flow created to validate and forward data to MongoDB Atlas.



3.6 MongoDB Charts configured for real-time dashboard visualization.